

CILOSOL 100

ANTI THROMBOTIC AGENT, PLATELET AGGREGATION INHIBITOR

Each tablet contains:

Cilostazol 100 mg

✧ **PRODUCT DESCRIPTION:**
White, round, flat-faced, bevelled edge tablet with engraved 100 on one side and plain on the other side

✧ **MECHANISM OF ACTION:**

Pharmacokinetics**Absorption**

Following multiple doses of Cilostazol 100 mg twice daily in patients with peripheral vascular disease, steady state is achieved within 4 days.

The C_{max} of Cilostazol and its primary circulating metabolites increase less than proportionally with increasing doses. However, the AUC for Cilostazol and its metabolites increase approximately proportionately with dose.

Biotransformation

The apparent elimination half-life of Cilostazol is 10.5 hours. There are two major metabolites, a dehydro-cilostazol and a 4'-trans-hydroxy cilostazol, both of which have similar apparent half-lives. The dehydro metabolite is 4-7 times as active a platelet anti-aggregant as the parent compound and the 4'-trans-hydroxy metabolite is one fifth as active. Plasma concentrations (as measured by AUC) of the dehydro and 4'-trans-hydroxy metabolites are ~41% and ~12% of Cilostazol concentrations. There is no evidence that Cilostazol induces hepatic microsomal enzymes.

Elimination

Cilostazol is eliminated predominantly by metabolism and subsequent urinary excretion of metabolites. The primary isoenzymes involved in its metabolism are cytochrome P-450 CYP3A4, to a lesser extent, CYP2C19, and to an even lesser extent CYP1A2. The primary route of elimination is urinary (74%) with the remainder excreted in the feces. No measurable amount of unchanged Cilostazol is excreted in the urine, and less than 2% of the dose is excreted as the dehydro-Cilostazol metabolite. Approximately 30% of the dose is excreted in the urine as the 4'-trans-hydroxy metabolite. The remainder is excreted as metabolites, none of which exceed 5% of the total excreted.

Distribution

Cilostazol is 95-98% protein bound, predominantly to albumin. The dehydro metabolite and 4'-trans-hydroxy metabolite are 97.4% and 66% protein bound respectively.

The pharmacokinetics of Cilostazol and its metabolites were not significantly affected by age or gender in healthy subjects aged between 50-80 years.

In subjects with severe renal impairment, the free fraction of Cilostazol was 27% higher and both C_{max} and AUC were 29% and 39% lower respectively than in subjects with normal renal function.

The C_{max} and AUC of the dehydro metabolite were 41% and 47% lower respectively in the severely renally impaired subjects compared to subjects with normal renal function. The C_{max} and AUC of 4'-trans-hydroxy cilostazol were 173% and 209% greater in subjects with severe renal impairment. The medicine must not be administered to patients with a creatinine clearance < 25 mL/min.

There are no data in patients with moderate to severe hepatic impairment and since Cilostazol is extensively metabolized by hepatic enzymes, the drug should not be used in such patients.

✧ **INDICATION*:**

Cilostazol is indicated for the improvement of the maximal and pain-free walking distances in patients with intermittent claudication, who do not have rest pain and who do not have evidence of peripheral tissue necrosis.

Prevention of recurrence of cerebral infarction (excluding cardiogenic cerebral embolism)

Precaution: The effects of Cilostazol on cerebral infarction have not been studied in patients with asymptomatic cerebral infarction.

✧ **DOSAGE AND ADMINISTRATION:**

Oral

The recommended dosage of Cilostazol is 100 mg b.i.d. taken at least half an hour before or two hours after breakfast and dinner. A dose of 50 mg b.i.d. should be considered during co-administration of such inhibitors of CYP3A4 as Ketoconazole, Itraconazole, Erythromycin and Diltiazem, and during co-administration of such inhibitors of CYP2C19 as Omeprazole.

Patients may respond as early as 2 to 4 weeks after the initiation of therapy, but treatment for up to 12 weeks may be needed before a beneficial effect is experienced.

Intermittent claudication

Cilostazol should be initiated by physicians experienced in the management of intermittent claudication.

The physician should reassess the patient after 3 months of treatment with a view to discontinuing Cilostazol where an inadequate effect is observed or symptoms have not been improved.

Patients receiving treatment with Cilostazol should continue with their life-style modifications (smoking cessation and exercise), and pharmacological interventions (such as lipid lowering and antiplatelet treatment) to reduce the risk of cardiovascular events.

Discontinuation of therapy: The available data suggest that the dosage of Cilostazol can be reduced or discontinued without rebound (i.e. platelet hyperaggregability).

✧ **WARNING AND PRECAUTION:**

The suitability of treatment with Cilostazol should be carefully considered alongside other treatment options such as revascularization. Based on its mechanism of action, Cilostazol may induce tachycardia, palpitation, tachyarrhythmia and/or hypotension. The increase in heart rate associated with Cilostazol is approximately 5 to 7 bpm; in patients at risk this consequently may induce angina pectoris. Patients who may be at increased risk for serious cardiac adverse events as a result of increased heart rate, e.g. patients with stable coronary disease, should be closely monitored during treatment with Cilostazol, while the use of Cilostazol in patients with unstable angina pectoris, or myocardial infarction/coronary intervention within the last 6 months, or a history of severe tachyarrhythmia is contraindicated. Caution should be exercised when prescribing Cilostazol for patients with atrial or ventricular ectopy and patients with atrial fibrillation or flutter.

Patients should be warned to report any episode of bleeding or easy bruising whilst on therapy. In case of retinal bleeding administration of Cilostazol should be stopped. Refer to Dosage and administration and Drug interaction for further information on bleeding risks.

Due to Cilostazol's platelet aggregation inhibitory effect it is possible that an increased bleeding risk occurs in combination with surgery (including minor invasive measurements like tooth extraction). If a patient is to undergo elective surgery and antiplatelet effect is not necessary, Cilostazol should be stopped 5 days prior to surgery.

There have been rare or very rare reports of hematological abnormalities including thrombocytopenia, leukopenia, agranulocytosis, pancytopenia and aplastic anemia. Most patients recovered on discontinuation of Cilostazol. However, some cases of pancytopenia and aplastic anemia had a fatal outcome.

In addition to reporting episodes of bleeding and easy bruising, patients should be warned to promptly report any other signs which might also suggest the early development of blood dyscrasia such as pyrexia and sore throat. A full blood count should be performed if infection is suspected or there is any other clinical evidence of blood dyscrasia. Cilostazol should be discontinued promptly if there is clinical or laboratory evidence of hematological abnormalities.

In the case of patients receiving strong inhibitors of CYP3A4 or CYP2C19, plasma levels of Cilostazol were shown to be increased. In such cases, a Cilostazol dosage of 50 mg twice daily is recommended (See Drug interaction for further information).

Caution is needed when co-administering Cilostazol with any other agent which has the potential to reduce blood pressure due to the possibility that there may be an additive hypotensive effect with a reflex tachycardia. Refer also to Side effect.

Caution should be exercised when co-administering Cilostazol with any other agents that inhibit platelet aggregation. Refer to Contraindication and Drug interaction.

✧ **PREGNANCY AND LACTATION:**

Pregnancy

There are no adequate data in the use of Cilostazol in pregnant women. The potential risk for humans is unknown. Cilostazol should not be used during pregnancy.

Lactation

The excretion of Cilostazol in human milk is unknown. Due to the potential harmful effect in the newborn child breastfed by a treated mother, the use of Cilostazol is not recommended during breastfeeding.

Fertility

The clinical significance is unknown.

✧ **SIDE EFFECT:**

The most common adverse reactions are headache, diarrhea and abnormal stools. These reactions were usually of mild to moderate intensity and were sometimes alleviated by reducing the dose.

Adverse reactions are included in the table below.

The frequencies of reactions which are considered unknown are those which cannot be estimated from the available data.

System Organ Class	Frequency	Side effects
Blood and the lymphatic system disorders	Common Uncommon Rare Unknown	Ecchymosis Anemia Bleeding time prolonged, thrombocythemia, Bleeding tendency, thrombocytopenia, granulocytopenia, agranulocytosis, leukopenia, pancytopenia, aplastic anemia
Immune system disorders	Uncommon	Allergic reaction
Metabolism and nutrition disorders	Common Uncommon	Edema (peripheral face), anorexia Hyperglycemia, diabetes mellitus
Psychiatric disorders	Uncommon	Anxiety
Nervous system disorders	Very common Common Uncommon Unknown	Headache Dizziness Insomnia, abnormal dreams Paresis, hypoaesthesia
Eye disorders	Unknown	Conjunctivitis
Ear and labyrinth disorders	Unknown	Tinnitus
Cardiac disorders	Common Uncommon	Palpitation, tachycardia, angina pectoris, arrhythmia, ventricular extrasystoles Myocardial infarction, atrial fibrillation, congestive heart failure, supraventricular tachycardia, ventricular tachycardia, syncope

Product	Code No.	Dimension	Packaging Type	Thickness
CILOSOL 100	ICMY 0071	W. 20 x L 26 cm.	Wood Free Paper (กระดาษปลอด)	60 g (0.08 mm.)
Designed by: (PDM, ASPD, PDCM, PDC-A, PDC-M, PDC-A)		Checked by: (PDM/ASPD)		Approved by: (MD (Only Pattern and Colors))
Approved by: PLC: (Date)	LRA: (Date)	IRA: (Date)	QCC-PM: (Date)	CUSTOMER/SALES DEPT.: (Date)

Vascular disorders	Uncommon Unknown	Eye hemorrhage, epistaxis, gastrointestinal hemorrhage, hemorrhage unspecified, orthostatic hypotension Hot flushes, hypertension, hypotension, cerebral hemorrhage, pulmonary hemorrhage, muscle hemorrhage, respiratory tract hemorrhage, subcutaneous hemorrhage
Respiratory, thoracic and mediastinal disorders	Common Uncommon Unknown	Rhinitis, pharyngitis Dyspnea, pneumonia, cough Interstitial pneumonia
Gastrointestinal disorders	Very common Common Uncommon	Diarrhea, abnormal feces Nausea and vomiting, dyspepsia, flatulence, abdominal pain Gastritis
Hepato-biliary disorders	Unknown	Hepatitis, hepatic function abnormal, jaundice
Skin and subcutaneous tissue disorders	Common Unknown	Rash, pruritus Eczema, skin eruptions, Steven's Johnson syndrome, toxic epidermal necrolysis, urticaria
Musculoskeletal, connective tissue and bone disorders	Uncommon	Myalgia
Renal and urinary disorders	Rare Unknown	Renal failure, renal impairment Hematuria, pollakiuria
General disorders and administration site conditions	Common Uncommon Unknown	Chest pain, asthenia Chills, malaise Pyrexia, pain
Investigations	Unknown	Uric acid level increased, blood urea increased, blood creatinine increased

✧ **EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:**
Cilostazol may cause dizziness and patients should be warned to exercise caution before they drive or operate machinery.

✧ **CONTRAINDICATION:**
• Known hypersensitivity to Cilostazol or to any of the excipients.
• Severe renal impairment: creatinine clearance of ≤ 25 mL/min
• Moderate or severe hepatic impairment
• Congestive heart failure
• Pregnancy
• Patients with any known predisposition to bleeding (e.g. active peptic ulceration, recent (within six months) hemorrhagic stroke, proliferative diabetic retinopathy, poorly controlled hypertension)
• Patients with any history of ventricular tachycardia, ventricular fibrillation or multifocal ventricular ectopics, whether or not adequately treated, and in patients with prolongation of the QTc interval.
• Patients with a history of severe tachyarrhythmia
• Patients treated concomitantly with two or more additional antiplatelet or anticoagulant agents (e.g. Acetylsalicylic acid, Clopidogrel, Heparin, Warfarin, Acenocoumarol, Dabigatran, Rivaroxaban or Apixaban).
• Patients with unstable angina pectoris, myocardial infarction within the last 6 months, or a coronary intervention in the last 6 months.

✧ **DRUG INTERACTION:**
Inhibitors of platelet aggregation
Cilostazol is a PDE III inhibitor with antiplatelet activity.
Acetylsalicylic Acid (ASA)
Short term (≤ 4 days) co-administration of ASA with Cilostazol suggested a 23-25% increase in inhibition of ADP-induced *ex vivo* platelet aggregation when compared to ASA alone.
Clopidogrel and other antiplatelet drugs
Concomitant administration of Cilostazol and Clopidogrel did not have any effect on platelet count, prothrombin time (PT) or activated partial thromboplastin time (aPTT). Caution is advised when co-administering Cilostazol with any drug that inhibits platelet aggregation. Consideration should be given to monitoring the bleeding time at intervals. Cilostazol treatment is contraindicated in patients receiving two or more additional antiplatelet/anticoagulant agents (See Contraindication).
Oral Anticoagulants like warfarin
Caution is advised in patients receiving both Cilostazol and any anticoagulant agent, and frequent monitoring is required to reduce the possibility of bleeding.
Cilostazol treatment is contraindicated in patients receiving two or more additional antiplatelet/anticoagulant agents (See Contraindication).
Cytochrome P-450 (CYP) enzyme inhibitors
Cilostazol is extensively metabolized by CYP enzymes, particularly CYP3A4 and CYP2C19 and to a lesser extent CYP1A2. The dehydro metabolite, which has 4-7 times the potency of Cilostazol in inhibiting platelet aggregation, appears to be formed primarily via CYP3A4. The 4'-trans-hydroxy metabolite, with potency one-fifth that of Cilostazol, appears to be formed primarily via CYP2C19. Therefore, drugs inhibiting CYP3A4 (e.g. some macrolides, azole antifungals, protease inhibitors) or CYP2C19 (like proton pump inhibitors, PPIs) increase the total pharmacological activity and could have the potential to enhance the undesirable effects of Cilostazol. Based on AUC, the overall pharmacological activity of Cilostazol increases 34% when co-administered with Erythromycin. Based on these data, the recommended dose of Cilostazol is 50 mg b.i.d. in the presence of erythromycin and similar agents (e.g. Clarithromycin).
Co-administration of Ketoconazole (an inhibitor of CYP3A4 with Cilostazol resulted in a 117% increase in the AUC of Cilostazol, accompanied by a 15% decrease in the AUC of the dehydro metabolite and a 87% increase in the AUC of the 4'-trans-hydroxy metabolite. Based on AUC, the overall pharmacological activity of Cilostazol increases 35% when coadministered with Ketoconazole. Based on these data, the recommended dose of Cilostazol is 50 mg b.i.d. in the presence of Ketoconazole and similar agents (e.g. Itraconazole).
Administration of Cilostazol with Diltiazem (a weak inhibitor of CYP3A4) resulted in an increase in the AUC of Cilostazol of 44%, accompanied by a 4% increase in AUC of the dehydro metabolite and a 43% increase in the AUC of the 4'-transhydroxy metabolite.
Based on AUC, overall pharmacological activity of Cilostazol increases 19% when co-administered with Diltiazem. Based on these data, no dose adjustment is necessary.
Administration of a single dose of 100 mg Cilostazol with 240 mL grapefruit juice (an inhibitor of intestinal CYP3A4) did not have a notable effect on the pharmacokinetics of Cilostazol. Based on these data, no dose adjustment is necessary. A clinically relevant effect on Cilostazol is still possible at higher quantities of grapefruit juice.
Administration of Cilostazol with Omeprazole (an inhibitor of CYP2C19) increased the AUC of Cilostazol by 22%, accompanied by a 68% increase in the AUC of the dehydro metabolite and a decrease of 36% in the AUC of the 4'-transhydroxy metabolite. Based on AUC, the overall pharmacological activity increases by 47% when co-administered with Omeprazole. Based on these data, the recommended dose of Cilostazol is 50 mg b.i.d. in the presence of Omeprazole.
Cytochrome P-450 enzyme substrates
Cilostazol has been shown to increase the AUC of Lovastatin (sensitive substrate for CYP3A4) and its β -hydroxy acid by 70%. Caution is advised when Cilostazol is co-administered with CYP3A4 substrates with a narrow therapeutic index (e.g. Cisapride, Halofantrine, Pimozide, ergot derivatives). Caution is advised in case of co-administration with statins metabolized by CYP3A4, for example Simvastatin, Atorvastatin and Lovastatin.
Cytochrome P-450 enzyme inducers
The effect of CYP3A4 and CYP2C19 inducers (such as Carbamazepine, Phenytoin, Rifampicin and St. John's wort) on Cilostazol pharmacokinetics has not been evaluated. The antiplatelet effect may theoretically be altered and should be carefully monitored when Cilostazol is co-administered with CYP3A4 and CYP2C19 inducers.
Other potential interactions
Caution is needed when co-administering Cilostazol with any other agent which has the potential to reduce blood pressure due to the possibility that there may be an additive hypotensive effect with a reflex tachycardia.

✧ **OVERDOSE AND TREATMENT:**
Information on acute overdose in humans is limited. The signs and symptoms can be anticipated to be severe headache, diarrhea, tachycardia and possibly cardiac arrhythmias. Patients should be observed and given supportive treatment. The stomach should be emptied by induced vomiting or gastric lavage, as appropriate.

✧ **SHELF LIFE:**
Three (3) years

✧ **STORAGE:**
Store at temperature of not more than 30°C.

✧ **DOSAGE FORM AND PACKAGING AVAILABLE:**
100 mg Tablet, Blister 10x10's

✧ **DATE OF REVISION:**
May 15, 2020

Manufactured by:
UNISON LABORATORIES CO., LTD.
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Product Registration Holder/ Importer:
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11400 Ayer Itam, Penang Malaysia

PM Specification	Product	Code No.	Dimension	Packaging Type	Thickness
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